



Matplotlib is a low level graph plotting library in python that serves as a visualization utility.

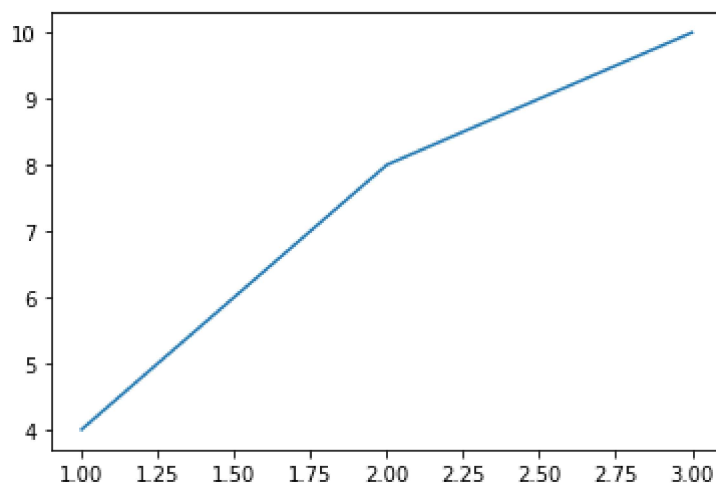
Matplotlib was created by John D. Hunter.

Matplotlib is open source and we can use it freely.

Matplotlib is mostly written in python, a few segments are written in C, Objective-C and Javascript for Platform compatibility.

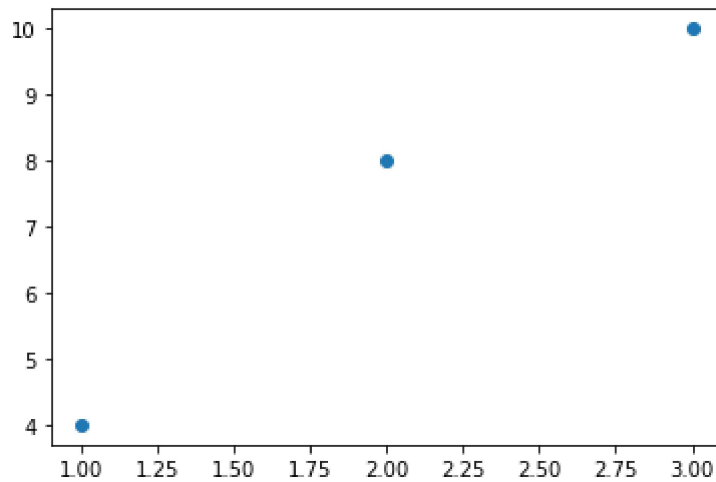
```
In [2]: from matplotlib import pyplot as plt  
import numpy as np
```

```
xpoints=np.array([1,2,3])  
ypoints=np.array([4,8,10])  
plt.plot(xpoints,ypoints)  
plt.show()
```



to plot without line just add o to plt.plot

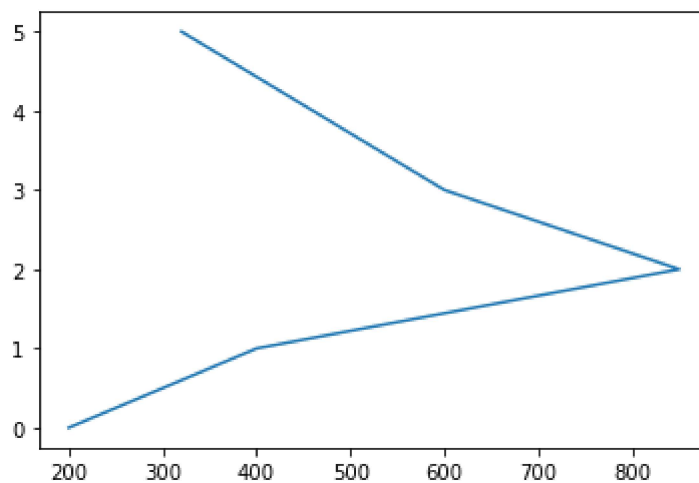
```
In [4]: plt.plot(xpoints,ypoints,'o')  
plt.show()
```



first we will cover all style aspects then we will se diff plot types

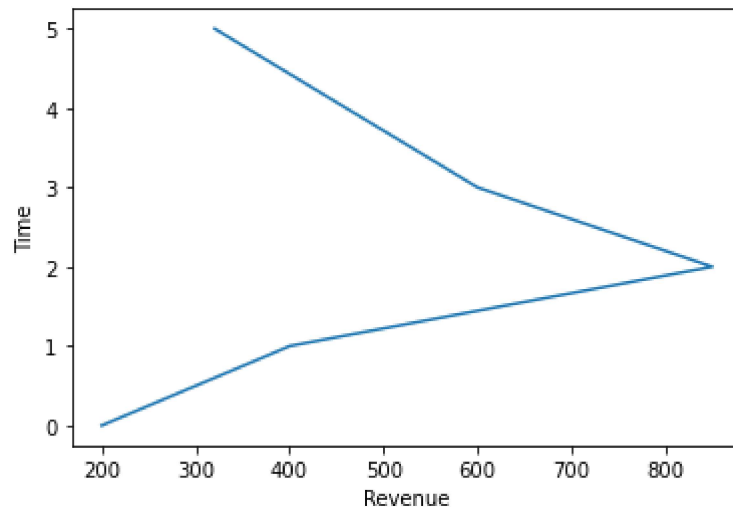
Style Aspects

```
In [6]: from matplotlib import pyplot as plt  
time=[0,1,2,3,5]  
revenue=[200,400,850,600,320]  
cost=[150, 500, 550, 550, 560]  
plt.plot(revenue,time)  
plt.show()
```



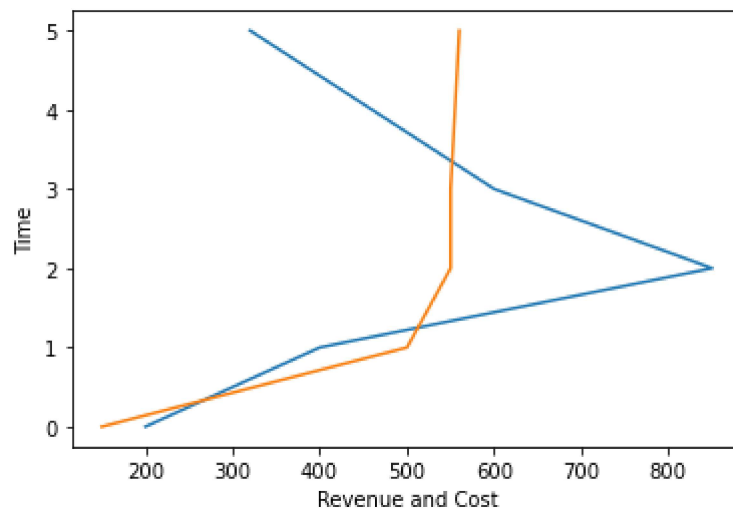
adding labels to x ,y axis

```
In [8]: plt.plot(revenue,time)
plt.xlabel('Revenue')
plt.ylabel('Time')
plt.show()
```



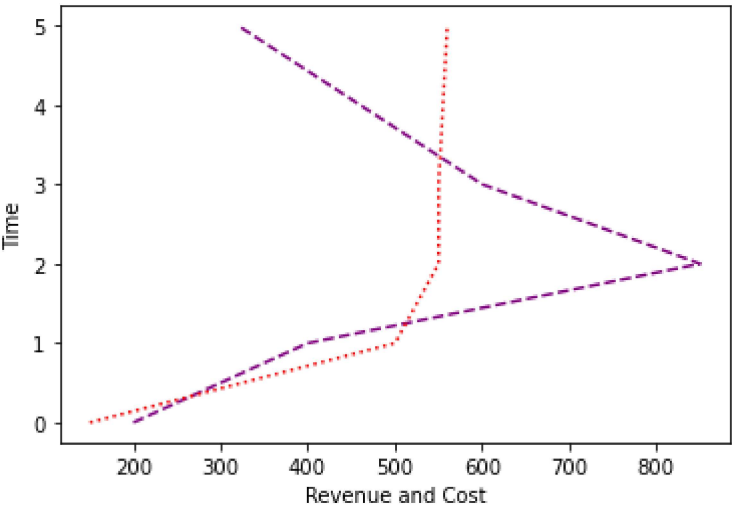
two graphs

```
In [10]: plt.plot(revenue,time)
plt.plot(cost,time)
plt.xlabel('Revenue and Cost')
plt.ylabel('Time')
plt.show()
```



now there are also diff line styles and colors

```
In [12]: plt.plot(revenue,time,color='purple',linestyle='dashed')
plt.plot(cost,time,color='red',linestyle='dotted')#color=#82edc9 binary
plt.xlabel('Revenue and Cost')
plt.ylabel('Time')
plt.show()
```



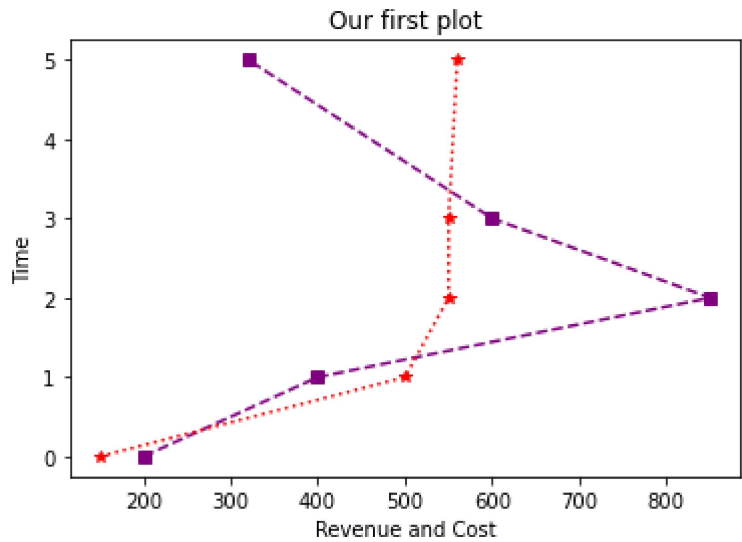
Line Styles

You can choose any of these styles:

Style	Or	
'solid' (default)	'-'	Try it »
'dotted'	'.'	Try it »
'dashed'	'--'	Try it »
'dashdot'	'-.'	Try it »
'None'	'' or ''	Try it »

there are also markers

```
In [17]: plt.plot(revenue,time,color='purple',linestyle='dashed',marker='s')
plt.plot(cost,time,color='red',linestyle='dotted',marker='*')#color=#82edc9 binary
plt.title('Our first plot')
plt.xlabel('Revenue and Cost')
plt.ylabel('Time')
plt.show()
```



Marker Reference

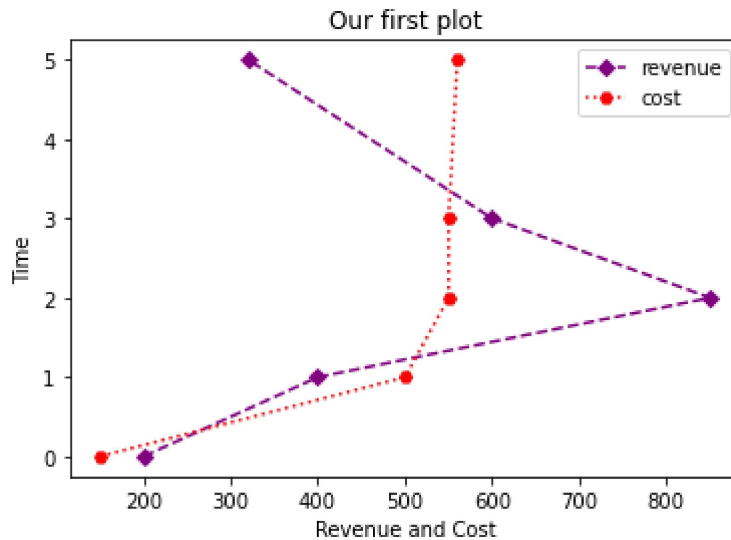
You can choose any of these markers:

Marker	Description
'o'	Circle
'*'	Star
'.'	Point
','	Pixel
'x'	X
'X'	X (filled)
'+'	Plus
'p'	Plus (filled)
's'	Square
'D'	Diamond
'd'	Diamond (thin)
'p'	Pentagon
'H'	Hexagon

adding legend

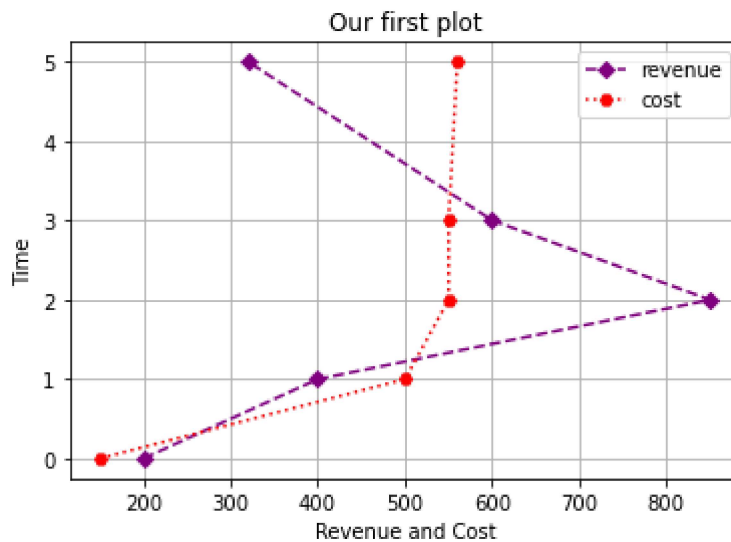
```
In [19]: plt.plot(revenue,time,color='purple',linestyle='dashed',marker='D')
plt.plot(cost,time,color='red',linestyle='dotted',marker='H')#color=#82edc9' binary
plt.title('Our first plot')
plt.xlabel('Revenue and Cost')
plt.ylabel('Time')
plt.legend(['revenue','cost'])

plt.show()
```



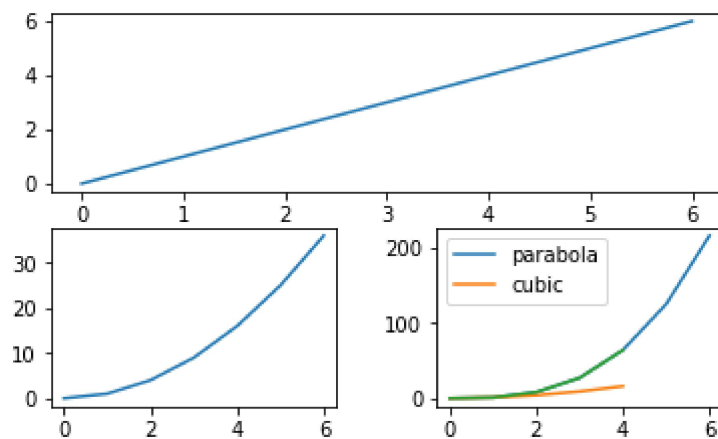
grid

```
In [20]: plt.plot(revenue,time,color='purple',linestyle='dashed',marker='D')
plt.plot(cost,time,color='red',linestyle='dotted',marker='H')#color=#82edc9' binary
plt.title('Our first plot')
plt.xlabel('Revenue and Cost')
plt.ylabel('Time')
plt.legend(['revenue','cost'])
plt.grid()
plt.show()
```



subplots

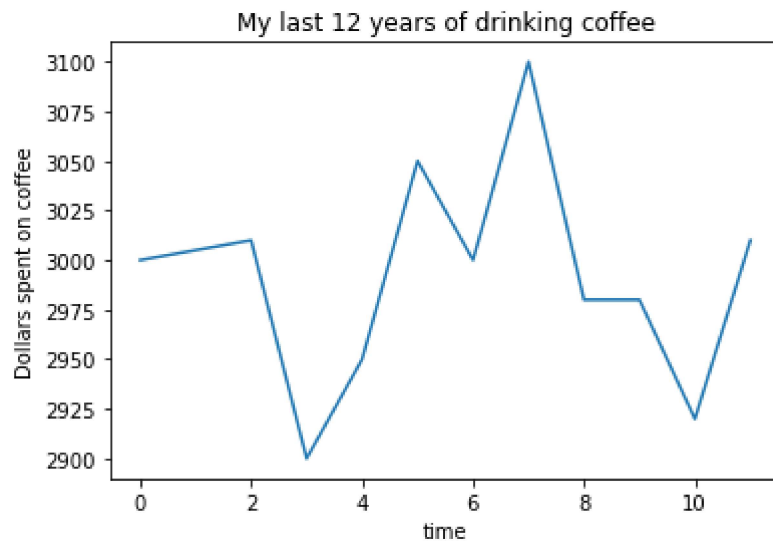
```
In [21]: from matplotlib import pyplot as plt
x = range(7)
straight_line = [0, 1, 2, 3, 4, 5, 6]
parabola = [0, 1, 4, 9, 16, 25, 36]
cubic = [0, 1, 8, 27, 64, 125, 216]
# Subplot 1
plt.subplot(2, 1, 1)
plt.plot(x, straight_line)
# Subplot 2
plt.subplot(2, 2, 3)
plt.plot(x, parabola)
# Subplot 3
plt.subplot(2, 2, 4)
plt.plot(x, cubic)
plt.subplots_adjust(wspace=0.35, bottom=0.2)
plt.plot([0, 1, 2, 3, 4], [0, 1, 4, 9, 16])
plt.plot([0, 1, 2, 3, 4], [0, 1, 8, 27, 64])
plt.legend(['parabola', 'cubic'])
plt.show()
```



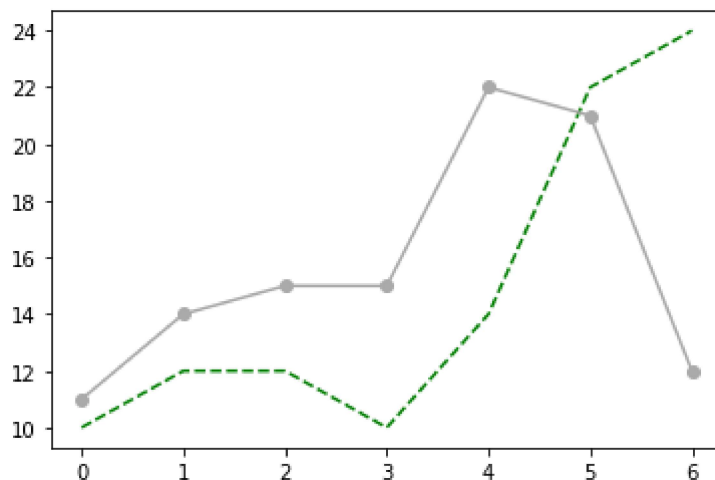
Diffrent Plot Types

Line plot

```
In [22]: from matplotlib import pyplot as plt
x = range(12)
y = [3000, 3005, 3010, 2900, 2950, 3050, 3000, 3100, 2980,
2980, 2920, 3010]
plt.title('My last 12 years of drinking coffee')
plt.xlabel('time')
plt.ylabel('Dollars spent on coffee')
plt.plot(x, y)
plt.show()
```



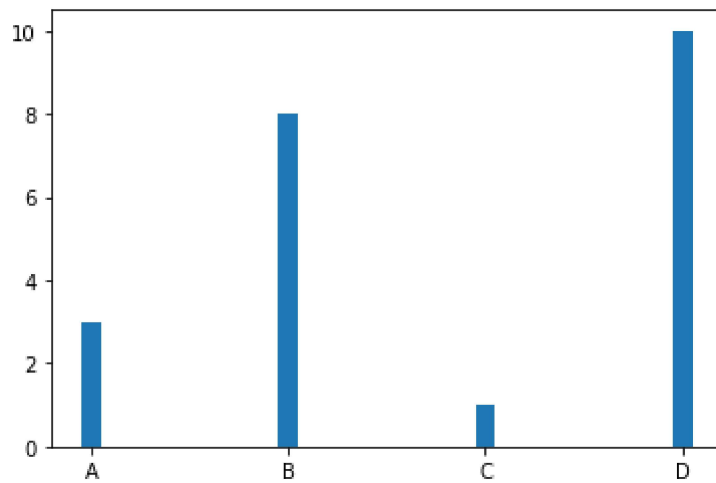
```
In [23]: from matplotlib import pyplot as plt
days = [0, 1, 2, 3, 4, 5, 6]
money_spent = [10, 12, 12, 10, 14, 22, 24]
money_spent_2 = [11, 14, 15, 15, 22, 21, 12]
plt.plot(days, money_spent, color='green', linestyle='--')
plt.plot(days, money_spent_2, color='#AAAAAA', marker='o')
plt.show()
```



Bar Plot

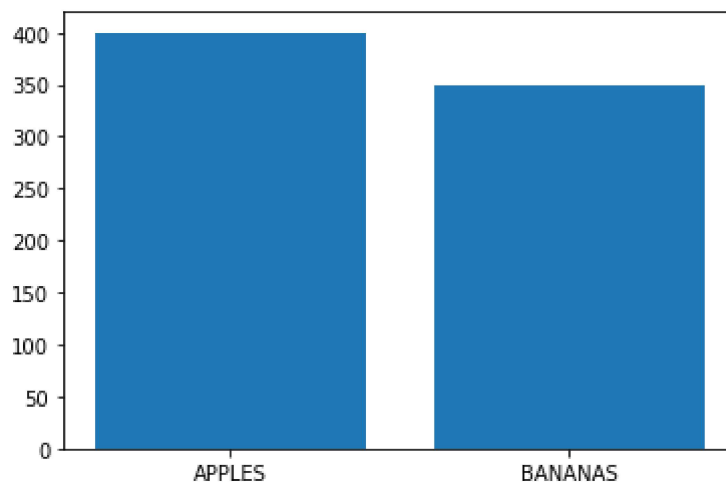
```
In [33]: from matplotlib import pyplot as plt
import numpy as np
x = np.array(["A", "B", "C", "D"])
y = np.array([3, 8, 1, 10])

plt.bar(x,y,width=0.1)
plt.show()
```



```
In [26]: x = ["APPLES", "BANANAS"]
y = [400, 350]
plt.show()
plt.bar(x, y)
```

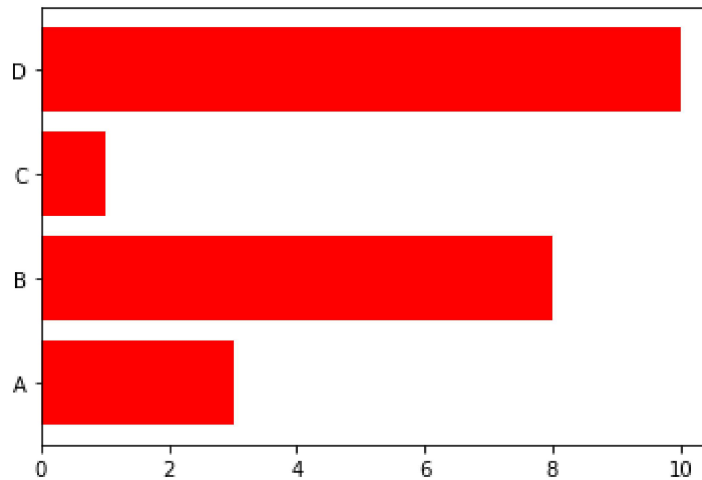
Out[26]: <BarContainer object of 2 artists>



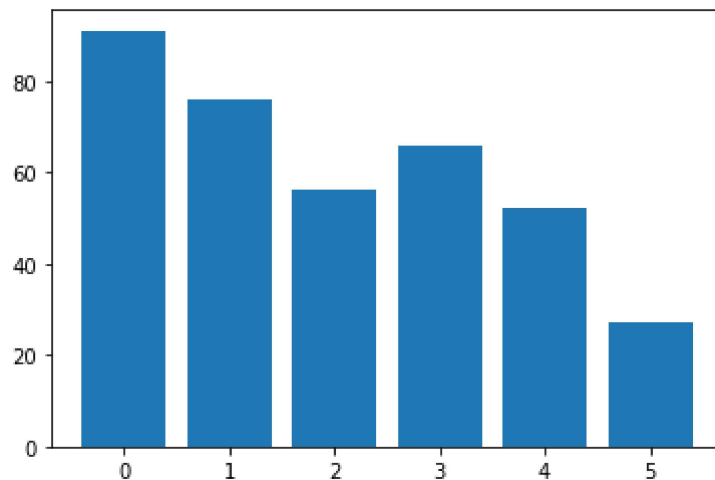
```
In [30]: # horizontal bar
import matplotlib.pyplot as plt
import numpy as np

x = np.array(["A", "B", "C", "D"])
y = np.array([3, 8, 1, 10])

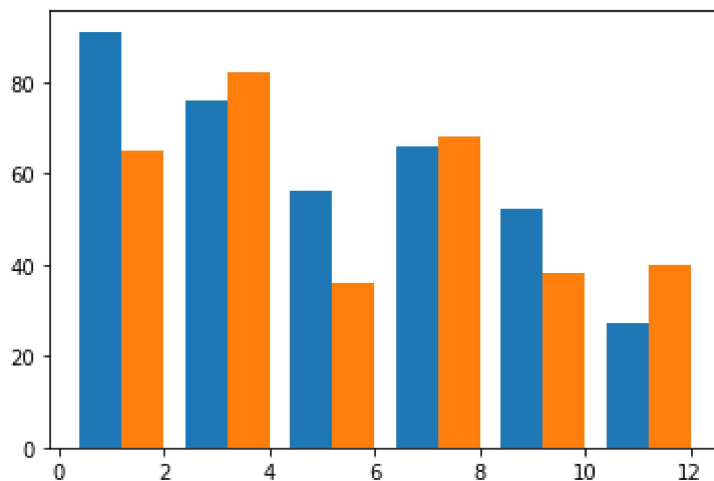
plt.barh(x, y, color='red')
plt.show()
```



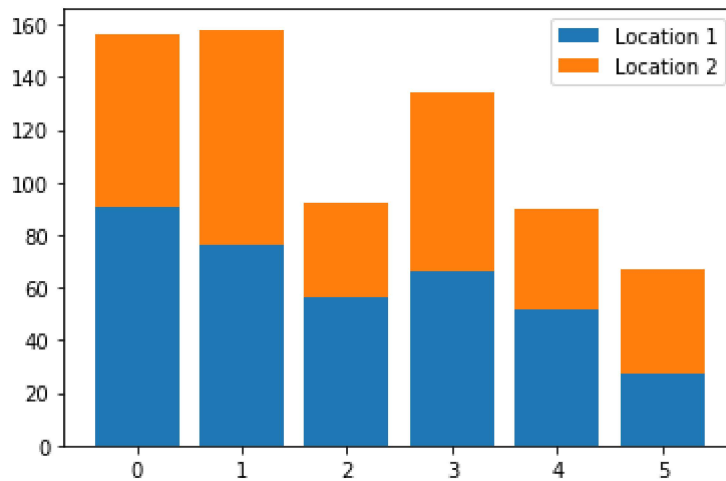
```
In [34]: from matplotlib import pyplot as plt
drinks = ["cappuccino", "latte", "chai", "americano", "mocha", "espresso"]
sales = [91, 76, 56, 66, 52, 27]
plt.bar(range(len(sales)), sales)
plt.show()
```



```
In [35]: from matplotlib import pyplot as plt
drinks = ["cappuccino", "latte", "chai", "americano",
"mocha", "espresso"]
sales1 = [91, 76, 56, 66, 52, 27]
sales2 = [65, 82, 36, 68, 38, 40]
#Paste the x_values code here
n = 1 # This is our first dataset (out of 2)
t = 2 # Number of dataset
d = 6 # Number of sets of bars
w = 0.8 # Width of each bar
store1_x = [t*element + w*n for element in range(d)]
plt.bar(store1_x, sales1)
#Paste the x_values code here
n = 2 # This is our second dataset (out of 2)
t = 2 # Number of dataset
d = 6 # Number of sets of bars
w = 0.8 # Width of each bar
store2_x = [t*element + w*n for element in range(d)]
plt.bar(store2_x, sales2)
plt.show()
```



```
In [36]: from matplotlib import pyplot as plt
drinks = ["cappuccino", "latte", "chai", "americano", "mocha",
"espresso"]
sales1 = [91, 76, 56, 66, 52, 27]
sales2 = [65, 82, 36, 68, 38, 40]
plt.bar(range(len(drinks)), sales1)
plt.bar(range(len(drinks)), sales2, bottom=sales1)
plt.legend(["Location 1", "Location 2"])
plt.show()
```



Histograms

```
In [37]: from matplotlib import pyplot as plt  
import numpy as np
```

```
# Creating dataset
```

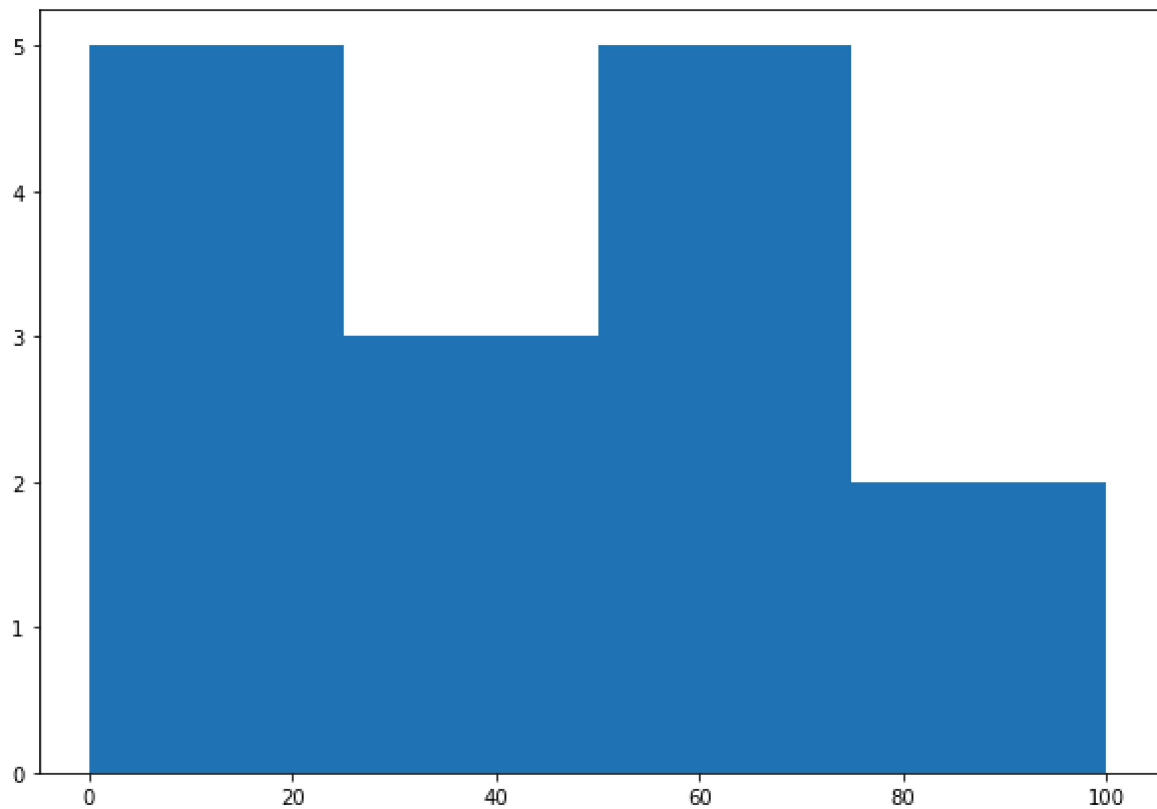
```
a = np.array([22, 87, 5, 43, 56,  
              73, 55, 54, 11,  
              20, 51, 5, 79, 31,  
              27])
```

```
# Creating histogram
```

```
fig, ax = plt.subplots(figsize=(10, 7))  
ax.hist(a, bins=[0, 25, 50, 75, 100])
```

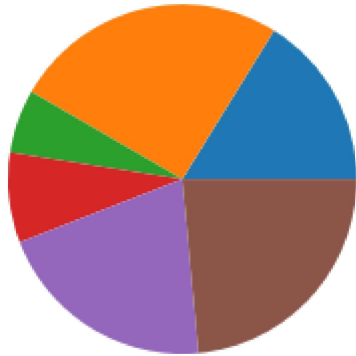
```
# Show plot
```

```
plt.show()
```

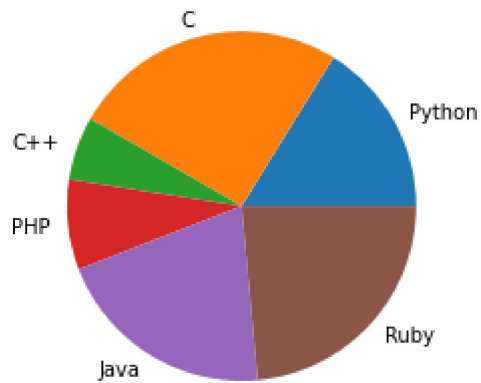


Piecharts

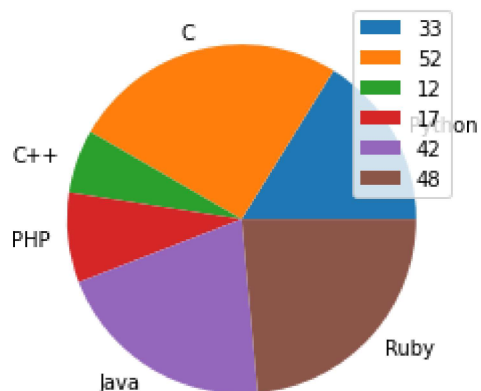
```
In [38]: from matplotlib import pyplot as plt  
data=[33,52,12,17,42,48]  
plt.pie(data)  
plt.show()
```



```
In [39]: labels=['Python','C','C++','PHP','Java','Ruby']  
plt.pie(data,labels=labels)  
plt.show()
```

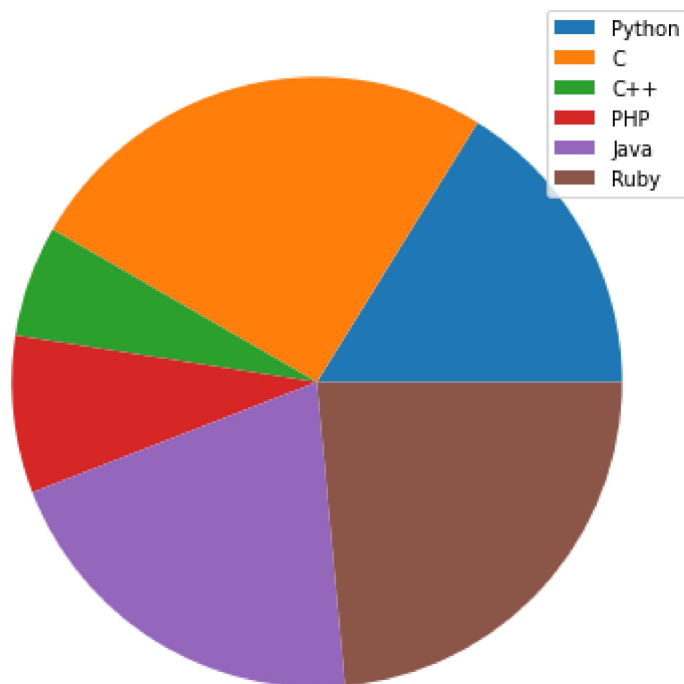


```
In [40]: plt.pie(data, labels=labels)
plt.legend(data)
plt.show()
```

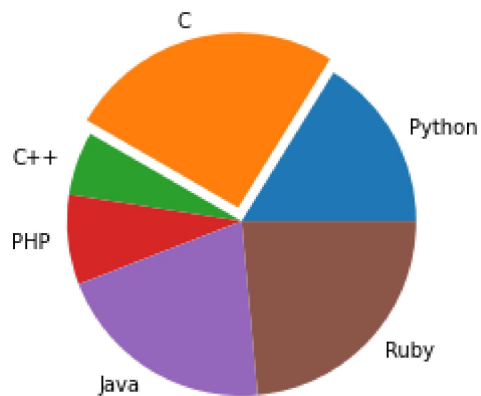


```
In [42]: plt.figure(figsize=(9, 7))
plt.pie(data)
plt.title('The results of the programming language survey', fontsize=14)
plt.legend(labels)
plt.show()
```

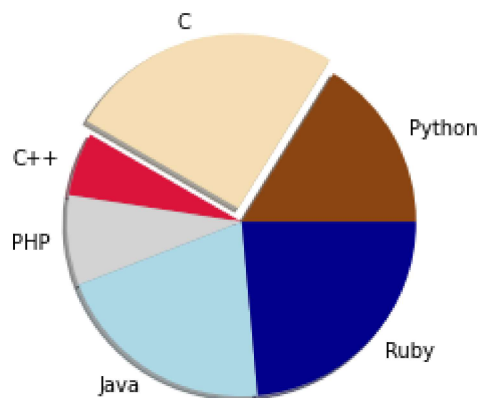
The results of the programming language survey



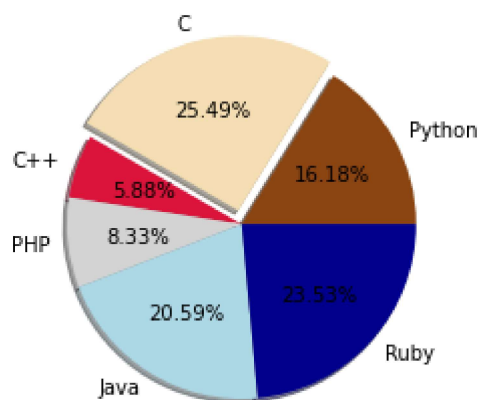
```
In [43]: # explode
explode = [0.0, 0.08, 0.0, 0.0, 0.0, 0.0]
plt.pie(data, explode=explode, labels=labels)
plt.show()
```



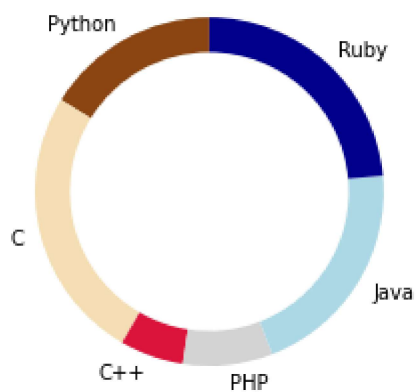
```
In [44]: colors = ['saddlebrown', 'wheat', 'crimson', 'lightgrey', 'lightblue', 'darkblue']
plt.pie(data,
        explode=explode,
        labels=labels,
        colors=colors,
        shadow=True)
plt.show()
```




```
In [45]: # display values
plt.pie(data,
        explode=explode,
        labels=labels,
        colors=colors,
        autopct='%0.2f%%', # here we also add the % sign
        shadow=True)
plt.show()
```



```
In [46]: plt.pie(data,
        labels=labels,
        colors=colors,
        startangle=90,
        wedgeprops={'width': 0.2})
plt.show()
```

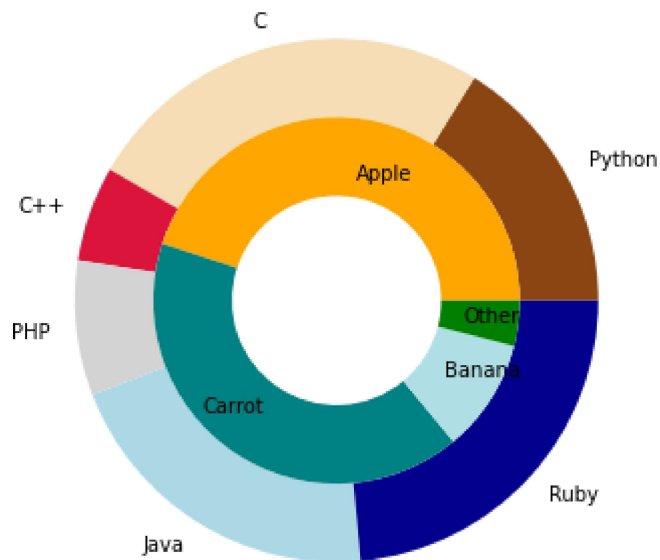


```
In [48]: fig, ax = plt.subplots(figsize=(8, 6))
width = 0.3

p_pie = ax.pie(data, radius=1, labels=labels, colors=colors, wedgeprops = {'width': width})

pets_data = [45, 41, 10, 4]
pets_labels = ['Apple', 'Carrot', 'Banana', 'Other']
pets_colors = ['orange', 'teal', 'powderblue', 'green']
pets_pie = ax.pie(pets_data,
                  radius=1-width,
                  labels=pets_labels,
                  labeldistance=0.7,
                  colors=pets_colors,
                  wedgeprops = {'width': width})

plt.show()
```

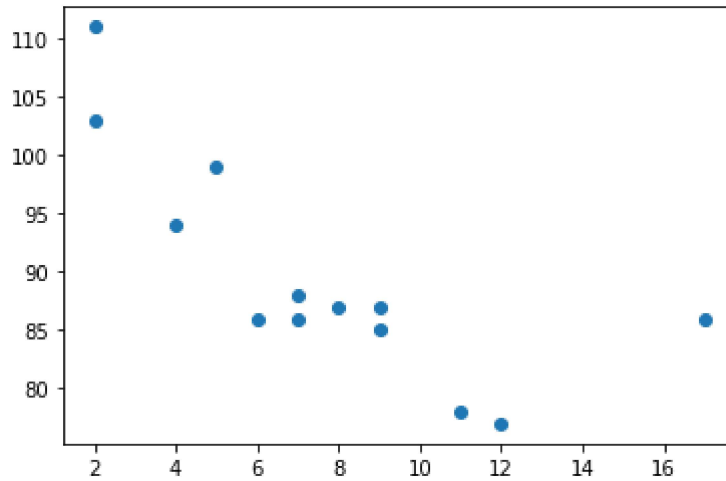


Scatter plot

```
In [52]: from matplotlib import pyplot as plt
import numpy as np

x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])

plt.scatter(x, y)
plt.show()
```

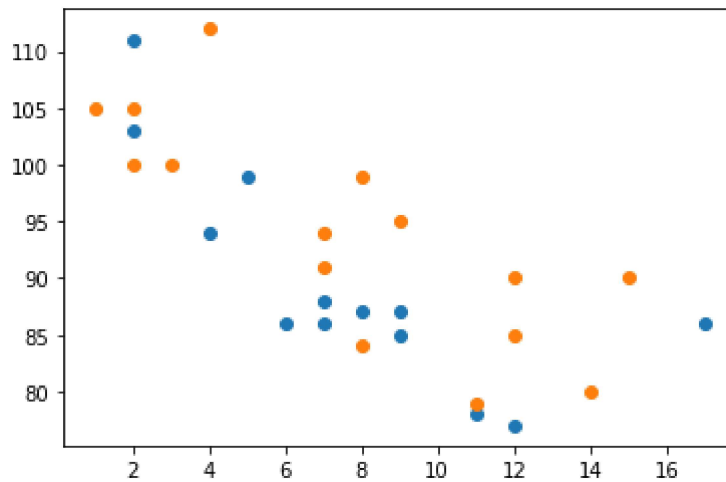


```
In [53]: from matplotlib import pyplot as plt
import numpy as np

#day one, the age and speed of 13 cars:
x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])
plt.scatter(x, y)

#day two, the age and speed of 15 cars:
x = np.array([2,2,8,1,15,8,12,9,7,3,11,4,7,14,12])
y = np.array([100,105,84,105,90,99,90,95,94,100,79,112,91,80,85])
plt.scatter(x, y)

plt.show()
```



Visualizing a dataset

```
In [54]: import pandas as pd
dataset=pd.read_csv('tips.csv')
dataset.head(10)
```

```
Out[54]:
```

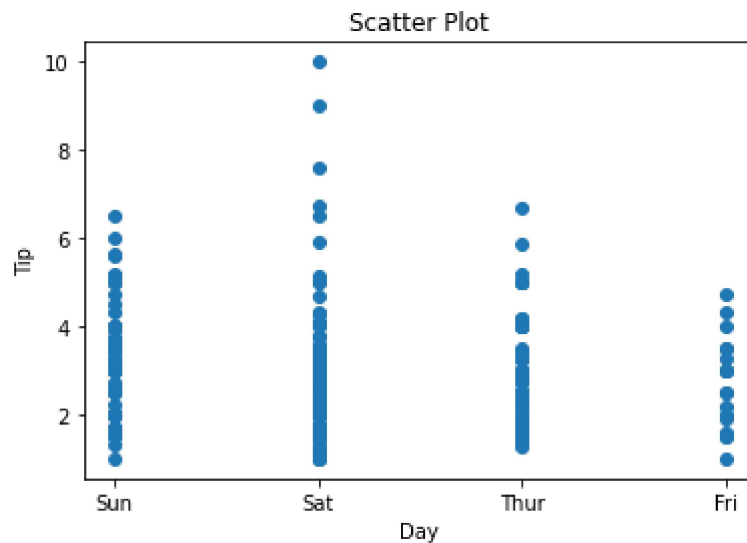
	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
5	25.29	4.71	Male	No	Sun	Dinner	4
6	8.77	2.00	Male	No	Sun	Dinner	2
7	26.88	3.12	Male	No	Sun	Dinner	4
8	15.04	1.96	Male	No	Sun	Dinner	2
9	14.78	3.23	Male	No	Sun	Dinner	2

```
In [55]: from matplotlib import pyplot as plt
plt.scatter(dataset['day'], dataset['tip'])

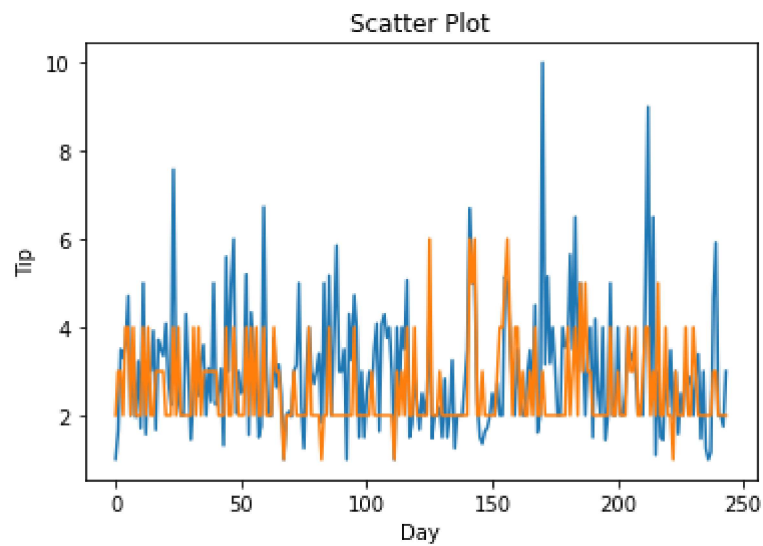
# Adding Title to the Plot
plt.title("Scatter Plot")

# Setting the X and Y labels
plt.xlabel('Day')
plt.ylabel('Tip')

plt.show()
```



```
In [56]: plt.plot(dataset['tip'])  
plt.plot(dataset['size'])  
  
# Adding Title to the Plot  
plt.title("Scatter Plot")  
  
# Setting the X and Y labels  
plt.xlabel('Day')  
plt.ylabel('Tip')  
  
plt.show()
```



```
In [62]: plt.bar(dataset['day'], dataset['tip'])
```

```
plt.title("Bar Chart")
```

```
# Setting the X and Y labels
```

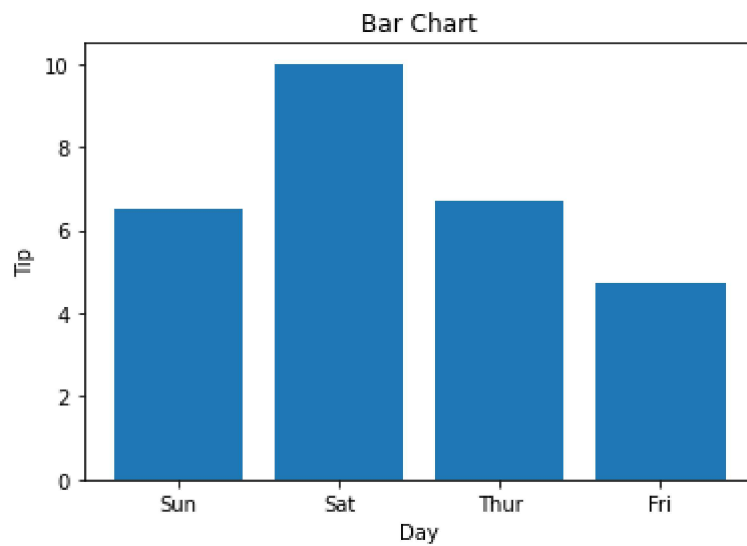
```
plt.xlabel('Day')
```

```
plt.ylabel('Tip')
```

```
# plt.legend('day', 'tip')
```

```
# Adding the legends
```

```
plt.show()
```

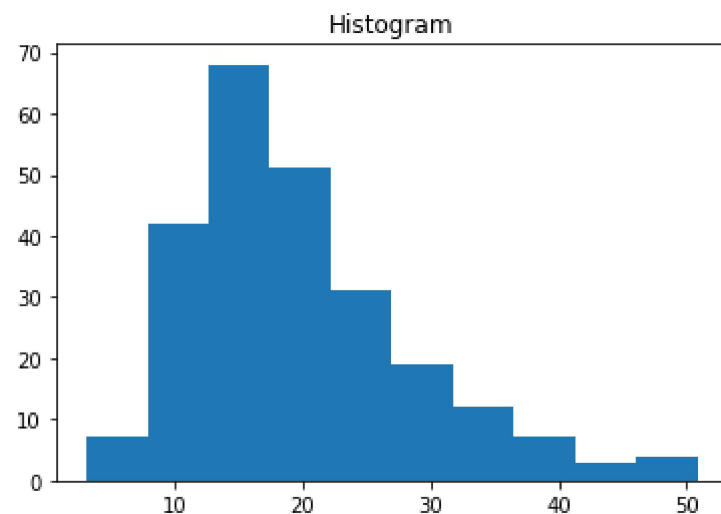


```
In [59]: plt.hist(dataset['total_bill'])
```

```
plt.title("Histogram")
```

```
# Adding the legends
```

```
plt.show()
```



Some Interesting stuff

```
In [64]: from mpl_toolkits.mplot3d import axes3d
from matplotlib import pyplot as plt
from matplotlib import style

style.use('fivethirtyeight')

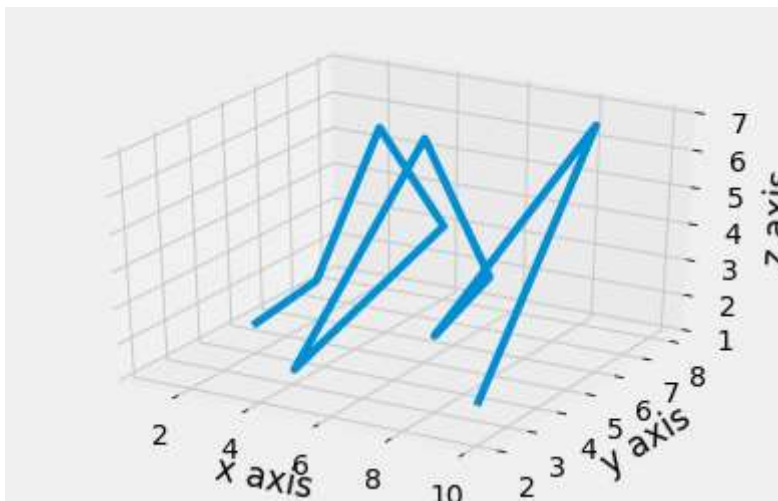
fig = plt.figure()
ax1 = fig.add_subplot(111, projection='3d')

x = [1,2,3,4,5,6,7,8,9,10]
y = [5,6,7,8,2,5,6,3,7,2]
z = [1,2,6,3,2,7,3,3,7,2]

ax1.plot(x,y,z)

ax1.set_xlabel('x axis')
ax1.set_ylabel('y axis')
ax1.set_zlabel('z axis')

plt.show()
```




```

In [65]: # 3D scatter
from mpl_toolkits.mplot3d import axes3d
from matplotlib import pyplot as plt
from matplotlib import style

style.use('ggplot')

fig = plt.figure()
ax1 = fig.add_subplot(111, projection='3d')

x = [1,2,3,4,5,6,7,8,9,10]
y = [5,6,7,8,2,5,6,3,7,2]
z = [1,2,6,3,2,7,3,3,7,2]

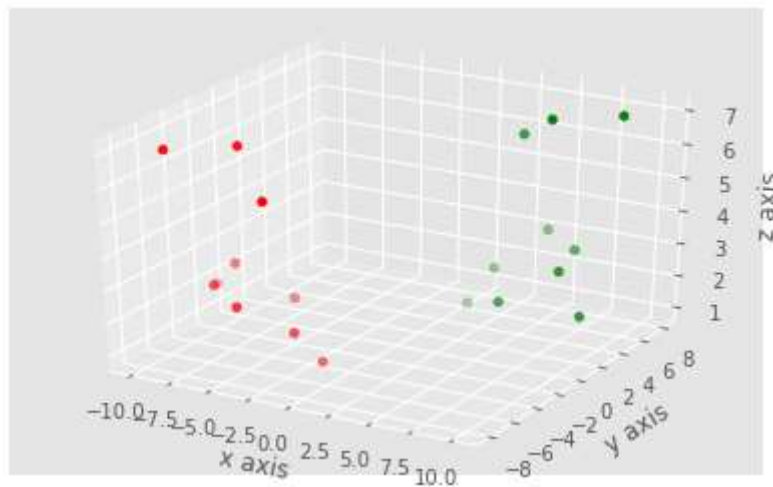
x2 = [-1,-2,-3,-4,-5,-6,-7,-8,-9,-10]
y2 = [-5,-6,-7,-8,-2,-5,-6,-3,-7,-2]
z2 = [1,2,6,3,2,7,3,3,7,2]

ax1.scatter(x, y, z, c='g', marker='o')
ax1.scatter(x2, y2, z2, c='r', marker='o')

ax1.set_xlabel('x axis')
ax1.set_ylabel('y axis')
ax1.set_zlabel('z axis')

plt.show()

```



```

In [66]: # 3D bar chart
from mpl_toolkits.mplot3d import axes3d
from matplotlib import pyplot as plt
import numpy as np
from matplotlib import style
style.use('ggplot')

fig = plt.figure()
ax1 = fig.add_subplot(111, projection='3d')

x3 = [1,2,3,4,5,6,7,8,9,10]
y3 = [5,6,7,8,2,5,6,3,7,2]
z3 = np.zeros(10)

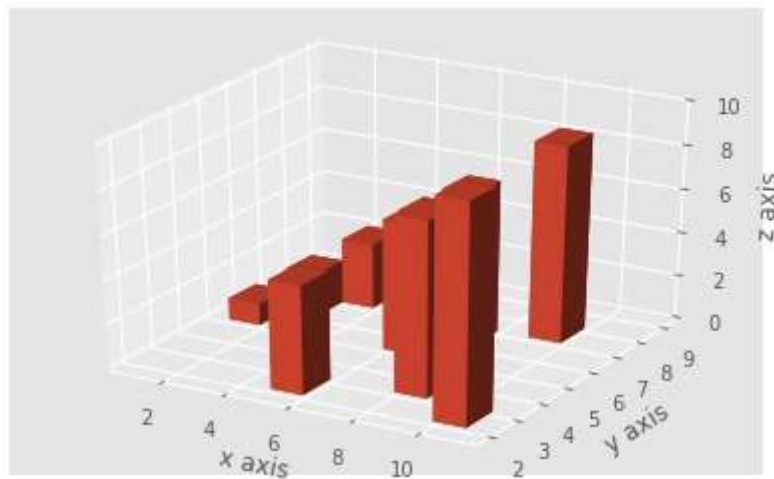
dx = np.ones(10)
dy = np.ones(10)
dz = [1,2,3,4,5,6,7,8,9,10]

ax1.bar3d(x3, y3, z3, dx, dy, dz)

ax1.set_xlabel('x axis')
ax1.set_ylabel('y axis')
ax1.set_zlabel('z axis')

plt.show()

```



Example 2

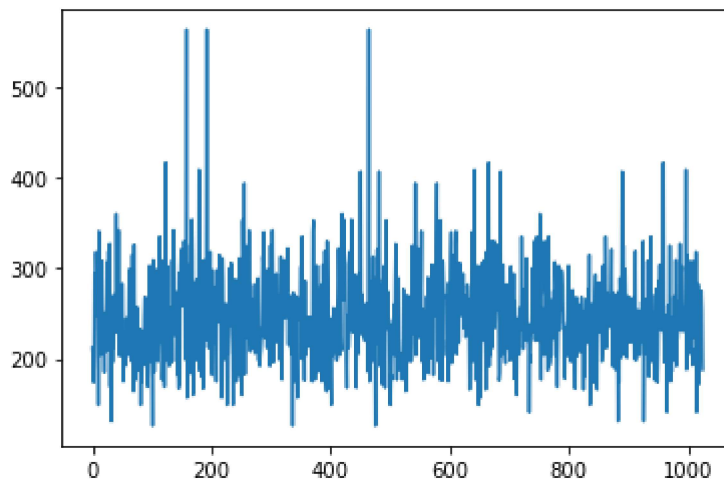
```
In [1]: import pandas as pd
from matplotlib import pyplot as plt
df2=pd.read_csv('heart.csv')
df2.head(10)
```

```
Out[1]:
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	52	1	0	125	212	0	1	168	0	1.0	2	2	3	0
1	53	1	0	140	203	1	0	155	1	3.1	0	0	3	0
2	70	1	0	145	174	0	1	125	1	2.6	0	0	3	0
3	61	1	0	148	203	0	1	161	0	0.0	2	1	3	0
4	62	0	0	138	294	1	1	106	0	1.9	1	3	2	0
5	58	0	0	100	248	0	0	122	0	1.0	1	0	2	1
6	58	1	0	114	318	0	2	140	0	4.4	0	3	1	0
7	55	1	0	160	289	0	0	145	1	0.8	1	1	3	0
8	46	1	0	120	249	0	0	144	0	0.8	2	0	3	0
9	54	1	0	122	286	0	0	116	1	3.2	1	2	2	0

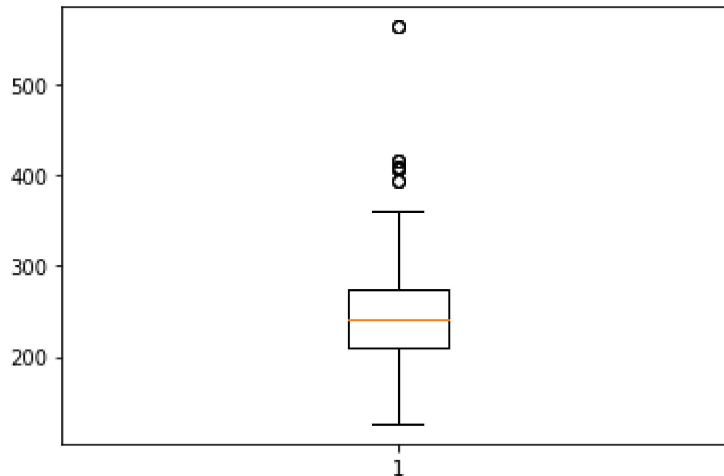
```
In [3]: plt.plot(df2['chol'])
```

```
Out[3]: [<matplotlib.lines.Line2D at 0x2eb04c5e430>]
```



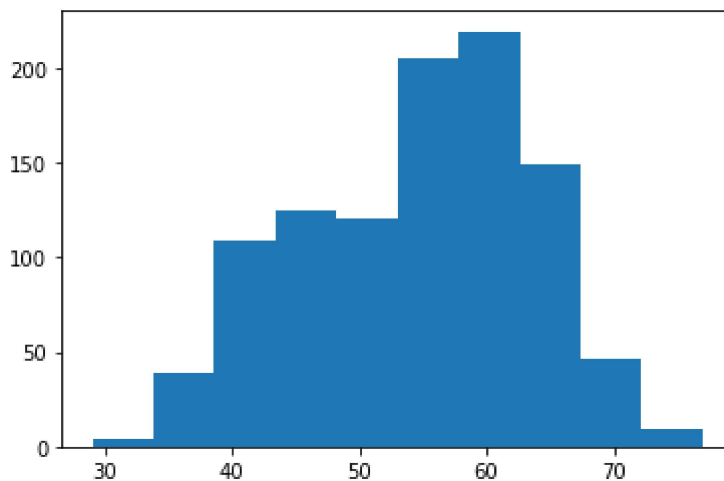
```
In [4]: plt.boxplot(df2['chol'])
```

```
Out[4]: {'whiskers': [<matplotlib.lines.Line2D at 0x2eb04cb3940>,  
  <matplotlib.lines.Line2D at 0x2eb04cb3ca0>],  
  'caps': [<matplotlib.lines.Line2D at 0x2eb04cc4040>,  
  <matplotlib.lines.Line2D at 0x2eb04cc43a0>],  
  'boxes': [<matplotlib.lines.Line2D at 0x2eb04cb35e0>],  
  'medians': [<matplotlib.lines.Line2D at 0x2eb04cc4700>],  
  'fliers': [<matplotlib.lines.Line2D at 0x2eb04cc4a00>],  
  'means': []}
```



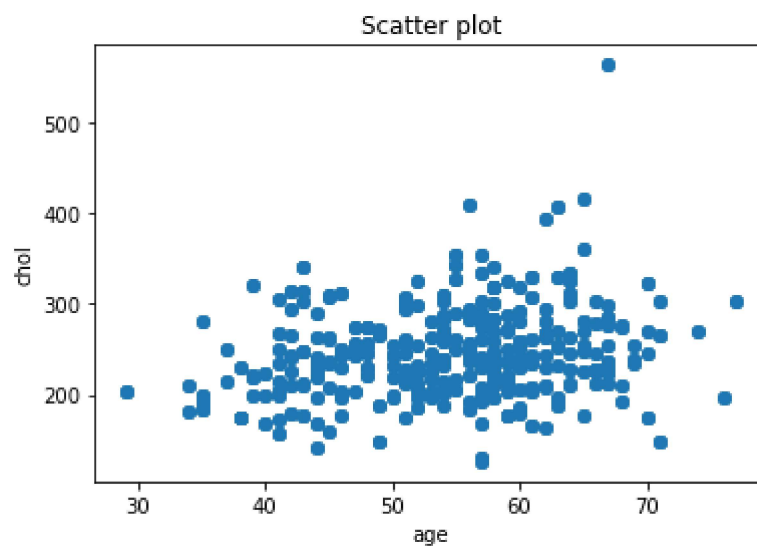
```
In [5]: plt.hist(df2['age'])
```

```
Out[5]: (array([ 4., 39., 109., 125., 120., 205., 219., 149., 46., 9.]),  
  array([29. , 33.8, 38.6, 43.4, 48.2, 53. , 57.8, 62.6, 67.4, 72.2, 77. ]),  
  <a list of 10 Patch objects>)
```



```
In [6]: plt.xlabel('age')  
plt.ylabel('chol')  
plt.title('Scatter plot')  
plt.scatter(x=df2['age'],y=df2['chol'])
```

Out[6]: <matplotlib.collections.PathCollection at 0x2eb04d7b7c0>



In [7]:

```
plt.figure(figsize=(15,10),dpi=100)
plt.xlabel('age')
plt.ylabel('random')
plt.legend()
plt.plot(df2['age'])
```

No handles with labels found to put in legend.

Out[7]: [

In []:

localhost:8888/notebooks/Desktop/DataScience_Python/DataVisualization_MATPLOTLIB.ipynb

30/30